
Mind, Market, and Mandate: A Field-Theoretic Synthesis for PPE

[T]-Theory Volume: Philosophy, Politics, and Economics

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Contents

Introduction: The Same Field Governs Mind, Market, and Mandate	4
Philosophy: Mind and the Field	4
Politics: Governance as Field Architecture	5
Economics: The Field as Market	5
The Patient Perspective	6
What This Book Offers the PPE Student or Scholar	6
The Programme	7
The Gap the Programme Addresses	7
The Structure of the Argument	8
The Method: Mathematical Co-identification	9
What It Is	9
Why It Matters as Method	9
When MCI Is Over: The Verification Threshold	10
The Model: The Soma-Field	12
Five Co-identifications	12
What the Model Predicts	13
The Empirical Test: QUANT-EXP-1	14
The Prediction	14
The Experiment	14
Results	14
The Penrose Connection	15
The Lived Case: Field Notes from the Inside	15
Extensions: Music, Film, and the Domain Generality of the Model	16
Music-Induced Affect	16
The Tensor: An Abstract Film	17
The Argument as a Whole	18
What Remains	19
Data and Code Availability	20
Introduction	21
Nash Equilibrium = Hopfield Minimum	21
Coordination Games and the Multi-Attractor Regime	22
Market Crashes as Phase Transitions	22
The FM-HN Extension: Regulatory Intervention	23
The Prisoner's Dilemma as Topological Obstruction	23
Applications	23
Conclusion	24

Introduction	25
Laws as Attractor Basin Constraints	25
Rights as Topological Invariants	25
Regulation as $J_{law}(t)$	26
The Rule of Law as Ergodicity	26
Legal Uncertainty and Attractor Fragmentation	27
Constitutional Law as Meta-Constraints	27
Applications and Predictions	28
Conclusion	28
A Note on Method	29
Introduction: The Inadequacy of Existing Maps	29
Background	30
Lived Experience as a Research Position	30
The Body-Mind Problem in Clinical Practice	31
The Felt Sense and Sub-Perceptual Emotion	31
Quantum Field Theory: Structure, Not Metaphor	31
Hopfield Networks and the Energy Function	32
The Soma-Field Model	32
Emotions as a Persistent Wave Field	32
The Perception Threshold	33
The Interaction of Emotional Modes	35
The Energy Landscape	35
The Hopfield Energy Function	35
Attractor States: Fight, Flight, Freeze, and Regulated Calm	35
Dissonance and Resolution	37
The Neurodivergent Field: ASD, ADHD, and C-PTSD as Operator Modifications	37
Complex PTSD: Memory Kernel and Asymmetric Coupling	37
ADHD: High Temperature, Low Damping, Pink Noise	38
Autism Spectrum Condition: Sparse Coupling and Modified Projection	38
The Composed Condition	39
The Soma-Field Instrument	39
Rationale	39
Design	39
The Feedback Loop	40
Pluggable Emotion Models	40
Clinical Implications	40
Assessment	40
Intervention	41
Psychoeducation	41

Neurodivergent Profiles as Structural Realities	41
Limitations and Future Directions	41
Conclusion	42

Introduction: The Same Field Governs Mind, Market, and Mandate

Oxford PPE — Philosophy, Politics, and Economics — is an unusual curriculum. It was assembled in 1920 with the explicit intention that the three disciplines illuminate each other: that philosophical rigour improves political reasoning, that economic analysis sharpens political choices, that political reality disciplines philosophical abstraction, that philosophical foundations constrain economic modelling. In practice, the three disciplines have increasingly gone their separate ways. Philosophy of mind has little contact with macroeconomics. Political theory debates justice in near-total isolation from game theory. Economists build equilibrium models without engaging the philosophical foundations of rational choice. The integrative promise of PPE has been partially fulfilled in applied ethics and political economy, but the deepest integration — a single framework that governs all three domains from first principles — has not been achieved.

This book proposes that the Universal Somatic Field (USF) provides exactly that framework. The same master equation that describes the dynamics of felt experience also describes Nash equilibrium selection in strategic games, the stability of legal and political institutions, and the dynamics of market systems. This is not a claim that philosophy, politics, and economics are *reducible* to physics. It is the claim that they all instantiate the same *mathematical structure* — a structure that the USF framework makes explicit and whose implications can be derived, not merely asserted.

Philosophy: Mind and the Field

The philosophical pillar of PPE begins with questions of mind, agency, and value. What is a preference? What makes an action rational? What is the good life, and what grounds its value? These are questions that standard economics treats as primitives — preference orderings are given, rationality is defined axiomatically, utility functions are postulated — and that philosophy finds the economic treatment unsatisfying precisely because it is circular: it defines rationality as preference-satisfaction and preferences as what rational agents have.

The USF framework grounds preferences in field dynamics: a preference is a gradient in the somatic field energy landscape. What you prefer is what your somatic field moves toward; what you disprefer is what it moves away from. Rationality, in this framework, is not an axiom but a property: a rational agent is one whose field dynamics track the gradient of a consistent energy landscape. Inconsistent preferences (the preference reversals documented by Kahneman and Tversky, the intransitivities observed in experimental economics) are not anomalies to be explained away; they are signatures of a landscape that has been modified by context — different attractors are accessible in different framings.

This gives the philosophy of mind a connection to the economics of decision-making that neither has had before. The phenomenology of choosing — the felt quality of desire, aversion, deliberation, regret — is the inside view of a field navigating its energy landscape. The normative question (what should one prefer?) is the question of what attractor landscape is worth having — a question that the framework makes well-posed even if it does not answer it.

Politics: Governance as Field Architecture

The political pillar of PPE addresses the organisation of collective life: justice, legitimacy, authority, democracy, rights. The USF framework connects directly to the accounts developed in the companion law volume: political institutions are constraint fields that modify the somatic field energy landscape; rights are topological invariants; the rule of law is ergodicity; legitimacy is the condition under which the constraint field is accepted as a modification of the landscape rather than resisted as an external imposition.

The specifically political question that the framework illuminates is the **design problem**: given a desired collective outcome (a fair distribution, a stable democracy, a sustainable commons), what institutional architecture — what pattern of attractors, barriers, and channels in the social somatic field — will produce that outcome as a stable equilibrium? This is a field-design problem, and the USF framework provides the tools to address it: the WKB formula for minimum intervention strength, the topological stability analysis for rights, the ergodicity criterion for the rule of law.

The framework also addresses the pathological cases. Authoritarian capture — the concentration of power in the hands of a single actor who modifies the landscape in their own interest — corresponds to the non-ergodic case: the powerful actor's field is protected from accountability by a deep, artificially maintained basin, while others are exposed to the full field dynamics. The field-theoretic description makes clear why authoritarian systems are unstable in the long run: they require continuous energy input to maintain the non-ergodic structure, and they become increasingly fragile to exogenous perturbations.

Economics: The Field as Market

The economic pillar connects directly to the companion economics volume: markets are somatic field systems, Nash equilibria are Hopfield minima, market crashes are phase transitions, and optimal regulation is a WKB problem. The distinctive contribution of the PPE synthesis is the connection between the economic and political/philosophical accounts.

Standard welfare economics evaluates market outcomes by their efficiency (Pareto optimality) and sometimes by their fairness (various social welfare functions). The USF framework adds a third criterion: **landscape robustness** — the degree to which the market attractor landscape is topologically stable against perturbations. An efficient-but-fragile market that crashes at the first shock may be worse, on landscape-robustness grounds, than a less efficient market with deep, stable attractors.

This connects economic analysis to the political design problem: the goal of economic regulation is not just to achieve efficient equilibria but to design a landscape that is topologically stable (rights-preserving), ergodic (rule-of-law satisfying), and efficient (welfare-maximising). The three criteria can conflict — there are trade-offs — but the framework provides a common vocabulary for navigating them.

The Patient Perspective

The patient perspective paper included in this volume serves a specific function in the PPE synthesis: it grounds the abstraction in a concrete human life. The field-theoretic account of trauma, recovery, and therapeutic intervention is not merely an application of the physics; it is a reminder of what the stakes of the inquiry are. Political institutions, economic systems, and legal frameworks all act on the somatic fields of real people. When they fail — when rights are violated, when markets crash and destroy livelihoods, when legal uncertainty leaves people unable to plan their lives — the failure has a phenomenal correlate: a somatic field driven into a trauma well, a field operating non-ergodically, a landscape fragmented by uncertainty.

The patient perspective paper makes this concrete. It is deliberately placed as the final chapter before the conclusion: the theory comes first, the lived experience comes last, and the juxtaposition is intentional.

What This Book Offers the PPE Student or Scholar

This volume is designed as the integrative text that the PPE curriculum has always needed but never had. Each chapter connects to the philosophical, political, or economic tradition it addresses; the field-theoretic framework is introduced through familiar PPE concepts rather than unfamiliar physics.

Chapter 2 (soma-field-synthesis) develops the field dynamics in terms accessible to the philosophically-trained reader. Chapter 3 (soma-game-theory, anchor paper) develops the economics and game theory. Chapter 4 (soma-law, anchor paper) develops the political and legal applications. Chapter 5 (soma-field-patient-pov) grounds the synthesis in lived experience. The final chapter addresses the integrative research agenda: what PPE questions the framework opens, and what interdisciplinary work is needed to pursue them.

Mind, market, mandate. The same equation, three vocabularies.

The Programme

This is a document about structure.

Not about feelings — though feelings are what the programme is ultimately for. Not about therapy — though therapy is one of the principal applications. Not about physics — though physics is where the mathematics comes from. It is about a single recurring observation: that the equations governing emotional dynamics are the same equations that govern quantum fields, and that this is not a metaphor.

When an identification like that is made precisely — when you can say not “this is *like* a wave” but “this *is* a wave in the technical sense, with the same propagator, the same energy function, the same topology, and therefore the same theorems” — a compressed body of work becomes possible. You are not building from scratch. You are navigating.

This document describes what was built by navigating, and why the pieces form a whole.

The Gap the Programme Addresses

Every large language model deployed today is a classical system. Its training is gradient descent. Its inference is deterministic or thermally noisy sampling. The architecture was designed to model the neocortex — pattern recognition, sequence prediction, error minimisation.

The complementary system — the limbic system, responsible for valuation, threat detection, arousal modulation, and the somatic state reinstatement that underlies trauma — had no formal mathematical treatment before this work. The clinical literature described it richly (Porges, van der Kolk, Levine). The neuroscience described its anatomy. Neither provided a model from which predictions could be derived and tested.

Simultaneously, the psychology of music had reached a similar ceiling. A 991-page handbook (Juslin and Sloboda, 2010) treated music-induced emotion almost entirely through Russell’s valence–arousal circumplex: a static two-dimensional map. The circumplex describes *where* a listener is, not *how* they move, what traps them, or what allows escape. No dynamical model of music-induced affect existed.

The programme fills both gaps with the same model, via the same method.

The Structure of the Argument

The argument has three movements and several extensions:

Paper	Movement	Contribution
<i>Mathematical Co-identification</i> (2026)	Method	Names and formalises the procedure
<i>The Soma-Field</i> (2026)	Model	Applies it to emotional dynamics
<i>Quantum Soma and the Penrose Gap</i> (2026)	Empirical test	Confirms the central claim
<i>Field Notes from the Inside</i> (2026)	Lived case	Primary-source clinical grounding
<i>A Dynamical Field Model of Music-Induced Affect</i> (2026)	Extension	Demonstrates domain generality
<i>The Tensor</i> (2026)	Extension	Applies the framework to abstract film

The popular account (*A Voyage into Trauma*, 2026) provides the same argument in accessible form, for readers without a physics background.

The Method: Mathematical Co-identification

What It Is

The history of mathematical science contains a recurring event. At a certain moment, a scientist recognises that the quantity they are studying is not *like* a quantity already understood in another domain — it *is* the same mathematical object, under a change of label. When this identification is made precisely, every theorem proved about the source object becomes available in the target domain immediately, without re-derivation.

This event has happened many times:

- Hopfield (1982) recognised that a neural network's energy-minimisation dynamics are the same as a spin-glass Hamiltonian. Every result from statistical mechanics of spin glasses — ground states, phase transitions, capacity bounds — imported.
- Veneziano (1968) recognised that the Euler Beta function, a result in pure mathematics, described the scattering amplitudes of hadrons. String theory began.
- Black and Scholes (1973) recognised that an option pricing equation was the heat diffusion equation. Every analytical tool from thermodynamics imported.

The paper *Mathematical Co-identification: A Method for Structural Import Across Scientific Domains* (Johnson, 2026a) names this procedure, formalises it as a distinct scientific method with its own validity criteria and failure modes, and distinguishes it from analogy, metaphor, and modelling. The key distinction:

Analogy: A is *like* B in certain respects. Illuminating, not transferable.

Co-identification: A *is* B under relabelling. Every theorem about B is a theorem about A.

Why It Matters as Method

A co-identification can be wrong. The identification is only valid if the mathematical type matches: the same dimensionality, the same algebraic structure, the same boundary conditions, the same symmetry group. The paper provides a falsifiability protocol — a formal procedure for pre-registering an import claim and specifying what observation would disconfirm it.

This matters because the failure mode of co-identification is not sloppy reasoning — it is overly precise reasoning applied to the wrong type. The paper catalogues seven historical examples to distinguish the valid from the invalid pattern.

The Soma-Field Model is the worked example throughout. The identification was not discovered by reading physics textbooks and looking for something that felt similar. It was discovered by writing down the equations the emotional system was observed to satisfy and recognising the form.

When MCI Is Over: The Verification Threshold

Here is something no paper on scientific method says clearly enough.

Every scientist who produces a major structural identification — Hopfield recognising the Ising Hamiltonian, Veneziano recognising the Euler Beta function — does so by being, for a period, a *type astronaut*. They are scanning the existing mathematical universe for a structure whose type signature matches the phenomenon they are studying. This is abductive reasoning: not deduction from first principles, not induction from data alone, but the systematic search of a known solution space for a structure that fits. It is, to be direct, a form of academic hacking. And it is how most of the connective work of mathematical science actually gets done.

What no one says — because scientists do not typically publish the search, only the result — is that the moment of structural identity is also the moment the method becomes irrelevant.

When Veneziano published his scattering amplitude, he did not need to cite the library where he found the Beta function. The formula was true. Every theorem about the Beta function was now a theorem about hadronic scattering. The library visit was the ladder; the amplitude was the building. The ladder came down.

The present work makes this transition explicit, because being explicit about it is itself a contribution. Every reader of the earlier papers — particularly *Mathematical Co-identification* — has correctly understood that MCI was the search engine. What should now be equally clear is that the search is over.

The exact moment the search ended was when the Lean 4 kernel accepted the proof. Not when a reviewer accepted a paper. Not when a human mathematician checked the algebra. When a formal proof-checking engine with no stake in the outcome closed the goal. That is the verification threshold. Before it: exploration. After it: structural fact.

What the work therefore rests on is not MCI. It rests on four things:

1. **Inductive structural necessity.** The 11-dimensional decomposition of a body-field-mind system is not an analogy with M-theory. It is the minimum geometry required to account for the functional degrees of freedom of a conscious organism. The isomorphism to M-theory is a *theorem*, not a design choice.
2. **Structural identity, not analogy.** A co-identification is not “A is like B.” It is “A *is* B under relabelling.” Every theorem about B becomes a theorem about A — immediately, without re-derivation. This is categorically different from saying that emotional dynamics *resemble* a Hopfield network. They *are* one.
3. **Scale invariance.** The same Helmholtz Green’s function equation governs 20 scales of physical reality, from quantum foam to the cosmic web. This is a discovered law. MCI was used to find it; it stands whether or not MCI is remembered.

4. **Kernel verification.** The Lean 4 proofs are the epistemological gold standard. A human reviewer can miss a subtle error. The kernel cannot. Kernel-verified theorems are exactly as true as the axioms they depend on — and the axioms are explicit and listed.

The *mathematical-co-identification* paper remains an accurate account of how the work was done — and naming the search process honestly is itself a contribution, in a discipline where the search is usually erased. But readers coming to the thesis or omnibus now should understand that they are reading the *results*, not the method. The method is in the history. The results are in the formal proofs.

The Model: The Soma-Field

Five Co-identifications

The Soma-Field Model (Johnson, 2026b) is built from five sequential co-identifications, each importing a body of mathematics from physics into emotional dynamics:

Co-identification 1: The Hopfield identification. The brain's emotional attractor dynamics satisfy the same energy function as a Hopfield neural network. The energy function is:

$$H(\mathbf{e}) = -\frac{1}{2}\mathbf{e}^\top W \mathbf{e} - \mathbf{b}^\top \mathbf{e}$$

where $\mathbf{e} \in \mathbb{R}^N$ is the emotional state vector, W is the coupling matrix, and $\mathbf{b}$ is a bias vector encoding baseline arousal. The local minima of H are the named attractor states: regulated calm, fight, flight, freeze, flow, dissociation.

Co-identification 2: The QFT identification. The emotional field propagates as a quantum field. The conscious emotional percept is the one-dimensional impulse response — the Green's function — of an eleven-dimensional coupling manifold. The same object that describes a massive particle in quantum field theory describes a conscious emotion: a pole in the propagator of the field.

$$G(\omega) = \frac{1}{\omega^2 - m^2 + i\epsilon}$$

This is not a metaphor. The threshold T at which a sub-perceptual field fluctuation becomes a conscious emotional percept is the mass parameter m in the propagator. Below threshold: virtual. Above threshold: real.

Co-identification 3: The brane identification. The body and the nervous system are not the same manifold. The body is a 3-brane embedded in the 11-dimensional coupling manifold. Somatic pain states and the body schema are field modes on this brane, not on the bulk manifold. This is the formal statement of the somatic grounding of emotion.

Co-identification 4: The G_2 holonomy identification. The seven compactified extra dimensions of the coupling manifold are a G_2 manifold. The G_2 holonomy group is the one that gives rise to topological obstructions — loops through the moduli space that cannot be continuously contracted to a point. In emotional terms: trauma configurations from which smooth continuous change cannot escape. The topological barrier is not a metaphor for being stuck. It is a mathematical object with a winding number.

Co-identification 5: The renormalisation group identification. Developmental trajectory maps onto the renormalisation group flow. The age at which a traumatic modification was introduced

corresponds to the energy scale at which the coupling constant was set. High-energy (early developmental) modifications are renormalisation-group relevant — they affect all subsequent scales. Low-energy (later life) modifications are irrelevant in the technical sense. This gives the formal account of why early trauma is not simply a more intense version of later trauma: it is a different class of object.

What the Model Predicts

From these five identifications, several predictions follow that are not derivable from any existing clinical model:

1. **Threshold crossings are phase transitions.** The transition from sub-perceptual to conscious emotion is a second-order phase transition in the field. This predicts hysteresis — it is easier to stay in a state than to enter it, and easier to stay out than to leave.
 2. **Complex PTSD is a topological configuration.** The coupling matrix W for a CPTSD nervous system has a specific structure: a winding-number-protected attractor landscape in which the Fear basin is separated from the Awe basin by a barrier that low-noise classical gradient descent cannot cross. This is a prediction about matrix structure, not a description of symptoms.
 3. **Autism Spectrum Condition modifies the threshold operator.** The threshold parameter T in the ASC nervous system has a different coupling to the field modes than in the neurotypical case — specifically, the threshold is non-uniform across sensory modalities, producing the characteristic pattern of simultaneous hypo- and hyper-sensitivity.
 4. **ADHD modifies the effective temperature.** The stochastic term in the Langevin dynamics governing the ADHD nervous system has higher effective temperature T_{eff} . This is not a deficit of attention; it is a higher rate of escape from local minima — an advantage in landscapes where rapid sampling is valuable and a liability where sustained convergence is required.
 5. **Quantum mechanisms are required for certain transitions.** For trauma configurations with topological barriers (non-zero winding number), low-noise classical gradient descent cannot reach the global minimum. A quantum mechanism is required. This is the prediction that QUANT-EXP-1 was designed to test.
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The Empirical Test: QUANT-EXP-1

The Prediction

The soma-field model makes a specific, falsifiable claim: for a Hopfield landscape with a topological trauma barrier, low-noise classical Langevin dynamics starting from the Fear attractor cannot reach the Awe attractor. Quantum annealing — a physically realisable mechanism — can.

This is not a claim about whether people should use quantum computers in therapy. It is a claim about reachability: that the mathematical structure of the barrier distinguishes the quantum and classical regimes in a measurable way.

The prediction was registered in the Zenodo v1 deposit of the Soma-Field paper (doi:10.5281/zenodo.20350515) before the experiment was run.

The Experiment

Quantum Soma and the Penrose Gap (Johnson, 2026c) reports QUANT-EXP-1: an exact 8-qubit statevector simulation on a 256-dimensional Hilbert space, implementing the Soma-Field Hopfield Hamiltonian with a transverse-field quantum annealing schedule.

The experimental design:

- **System:** 8-qubit Hopfield model encoding four emotional modes (Fear, Calm, Awe, Grief) plus sub-modes. Coupling matrix W set to produce a topological barrier between Fear and Awe.
- **Quantum dynamics:** Transverse-field annealing with schedule $H(s) = (1-s)H_X + sH_{\text{problem}}$, $s \in [0, 1]$.
- **Classical baseline:** Overdamped Langevin dynamics at low temperature ($T_{\text{eff}} = 0.01$), same starting state, same landscape.
- **Primary outcome:** Peak Awe-dominant occupancy (quantum) versus success rate of cold-classical crossings.

Results

Results are presented against the pre-registered barrier ladder:

Barrier strength	Classical cold rate	Classical cold CI	
		[95%]	Quantum peak
$W = -6$	0.000	[0.000, 0.019]	0.389
$W = -8$	0.000	[0.000, 0.019]	0.408
$W = -10$	0.000	[0.000, 0.019]	0.408
$W = -12$	0.000	[0.000, 0.019]	0.409
$W = -14$	0.000	[0.000, 0.019]	0.416

Bootstrap confidence intervals ($n = 200$ seeds) confirm that the classical cold success rate is bounded above by 1.9% at all tested barrier strengths. Quantum peak occupancy is stable at 0.389–0.416 across the full range.

Pre-registered hardening protocol — all checks passed:

- **Bootstrap** ($n = 200$): cold CI = [0.000, 0.019]; quantum peak 0.408–0.410. Intervals do not overlap at any barrier strength.
- **Control A** (start from Awe, barrier intact): classical 16/16 stay in Awe. PASS. Confirms that the barrier is directional: it blocks Fear $\rightarrow$ Awe, not the reverse.
- **Control B** (barrier removed, $W[\text{Fear}, \text{Awe}] = +0.4$): classical 16/16 reach Awe. PASS. Confirms that the barrier, not the landscape geometry, is what blocks classical dynamics.
- **Spectral gap**: gap narrows monotonically with barrier strength (B8: 0.0095, B10: 0.0089, B12: 0.0085) and reaches its minimum at $s \approx 0.999$, confirming the tunnelling bottleneck is late in the anneal.

Verdict: The strong reachability claim stands. QUANT-EXP-1 is a PASS.

The Penrose Connection

The paper situates this result in the context of Penrose’s argument about non-computability and consciousness. The connection is not that consciousness requires quantum mechanics in general. The connection is more specific:

Penrose identified a *gap* between what classical computation can reach and what consciousness can do. The soma-field identifies a corresponding *topological gap* in the emotional landscape between what classical gradient descent can reach and what a genuinely new state of the nervous system requires. QUANT-EXP-1 provides the computational demonstration that the gap exists and is crossable by a quantum mechanism.

The contribution is not to resolve Penrose’s claim about consciousness. It is to *instantiate* the gap in a concrete, testable, mathematical setting.

The Lived Case: Field Notes from the Inside

Field Notes from the Inside: A Patient-Constructed Model of Emotional Dynamics (Johnson, 2026d) performs a function that the formal papers cannot perform: it provides the primary-source clinical grounding.

The paper is written by the person who has Autism Spectrum Condition (Level 2), Attention Deficit Hyperactivity Disorder, and Complex Post-Traumatic Stress Disorder — and who also has a degree in physics. The model was not developed by observing patients. It was developed by having the conditions and finding the existing models inadequate.

The epistemological contribution of this paper is often undervalued. Every formal model of a human system is, in the end, derived from observation of that system. When the observer and the observed are the same entity, and that entity has the training to translate observation into formal mathematics, the resulting model has a different epistemic status from one derived by observation from the outside. The paper makes this explicit, situates it within the autoethnographic research tradition, and argues that the resulting model is *more* constrained, not less — because any prediction the model makes that does not match the primary observer’s experience is immediately falsified.

The formal content is a set of operator modifications for the three conditions:

- **ASC:** The threshold operator T is replaced by a modality-dependent operator T_k for each sensory channel k , with different coupling strengths. The result is the characteristic simultaneous hypo- and hyper-sensitivity: some channels are below threshold where the neurotypical channel is above it, others are above where the neurotypical channel is below.
- **ADHD:** The Langevin noise term $\sqrt{2T_{\text{eff}}}\eta(t)$ has elevated T_{eff} . This is a quantitative modification, not a qualitative one. The system is not broken; it is sampling the energy landscape at higher temperature. The therapeutic implication is not to reduce the noise but to design the landscape so that high-temperature sampling is an advantage.
- **CPTSD:** The coupling matrix W has the topological structure described in §3.2: a winding-number-protected barrier between Fear and regulated states. The barrier was installed before language, before narrative memory, before the self that can explain the barrier was formed. The modification is not a layer added to a pre-existing structure. It is the structure.

Extensions: Music, Film, and the Domain Generality of the Model

Music-Induced Affect

A Dynamical Field Model of Music-Induced Affect: Beyond the Valence–Arousal Circumplex (Johnson, 2026e) applies the soma-field framework to a domain where the empirical literature is rich and the theoretical models are weak.

Juslin and Sloboda’s *Handbook of Music and Emotion* (2010) — 991 pages — contains the circumplex as its dominant quantitative framework. The circumplex is a static map. It describes where a listener is; it does not model how they move. The soma-field is the first dynamical model of music-induced affect.

The key predictions that the circumplex cannot make but the field model does:

1. **Phase transitions, not continuous shifts.** State changes in music-induced affect are not smooth movements across the circumplex. They are threshold crossings — sudden re-configurations

of the attractor landscape. The field model predicts the conditions under which a transition occurs and the hysteresis that prevents immediate return.

2. **The adaptive function of high effective temperature.** In the ADHD nervous system (elevated T_{eff}), music that holds a neurotypical listener in a stable state may drive repeated transitions. This is not a bug; it is the same high sampling rate that characterises the ADHD cognitive profile. The model gives this a formal account.
3. **Basin depth asymmetry.** The freeze attractor basin is deeper than the regulated calm basin. This means it is harder to leave freeze than it is to leave calm — asymmetric with respect to the direction of transition. Music that successfully moves a listener from freeze to calm is doing qualitatively different work than music that moves a calm listener to a more activated state.

The paper also specifies a real-time instrument implementation: a MIDI controller array driving a Python field server at 50 Hz, with audio output via Ableton Live and 3D fractal visual output (Mandelbulb projection onto HoloGauze screen). The instrument is not described; it is specified formally, with pre-registered hypotheses and disconfirmation criteria.

The Tensor: An Abstract Film

The Tensor: An Abstract Film Definition (Johnson, 2026f) extends the framework to abstract film. A film is defined not by its pixels but by its **emotional score**: a vector-valued trajectory $\mathbf{e}^*(t)$ through the emotional field, parameterised by story-time $t \in [0, 1]$.

The rendering — the actual audiovisual output a viewer experiences — is generated at runtime from this trajectory, the viewer's own soma-field state, and a set of control parameters. In the limit where the viewer's biofeedback is available, the film adapts to where the viewer is: the trajectory is not what the viewer experiences, but what the film proposes. The work is not the pixels. It is the map.

This is a significant claim about what an artwork is. A conventional film is fixed: the same sequence of frames for every viewer at every screening. The tensor film is a field: a mathematical object that takes the viewer's state as input and produces an output adapted to it. The artistic statement is in the trajectory, not the realisation.

The paper does not describe how to make such a film. It defines the abstract structure that any realisation of such a film must instantiate — the way a musical score defines a symphony without being the performance.

The Argument as a Whole

The six papers form a single argument, and it can be stated in a paragraph:

The limbic system and its coupling to the body are governed by the same mathematical equations as a quantum field on a manifold with G_2 holonomy. This identification is not a metaphor; it is a co-identification in the technical sense, with all the theorems of each source domain importing into the target. Among those theorems is one that has clinical consequences: topological barriers in the emotional attractor landscape cannot be crossed by low-noise classical gradient descent. A quantum mechanism can cross them. This has been computationally confirmed (QUANT-EXP-1) against a pre-registered hardening protocol. The model correctly describes the structure of Autism Spectrum Condition, ADHD, and Complex PTSD as operator modifications, and generalises to music-induced affect and abstract film with no change to the underlying mathematics.

What makes this a research programme rather than a single paper is the **generativity**: the method (co-identification) produces results in any domain where an attractor landscape with topological structure can be identified. The soma-field is one instantiation. Music-induced affect is a second. Abstract film is a third. A fourth — currently in design — is **H-AL**: a holographic avatar whose body is a live Mandelbulb rendering of the emotional field state, projected at human scale through a hologauze screen and accompanied by a synthesised voice narrating the field in real time. The geometry of the fractal changes as the field changes; regulated calm and trauma produce visually distinct and mathematically characterisable forms. The same functor architecture (§A.4 of the main paper) supports this output with no changes to the field computation. Each of these instantiations generates falsifiable predictions from the same mathematical core.

What makes this a *novel* research programme is the **gap it fills**: no formal dynamical model of the limbic system existed before this work. The Hopfield framework gave the neocortex its formal model in 1982. The soma-field gives the limbic system its formal model in 2026. Together they constitute the first complete formal description of the two principal computational substrates of the vertebrate brain.

What Remains

The body of work described here is computationally complete. All pre-registered hardening checks have been executed. The claims that can be confirmed by simulation have been confirmed.

Three categories of work remain outside the scope of these papers:

Physical hardware confirmation. QUANT-EXP-1 uses exact statevector simulation. Running the same 8-qubit experiment on IBM Quantum free-tier hardware would produce the sentence “confirmed on physical quantum hardware.” This is feasible, is the logical next step for any journal submission targeting a hardware-inclusive venue, and is not required to support any claim in the current corpus.

Peer review. The three published papers are currently archived on Zenodo as open preprints. Peer review in ranked journals is a separate track, ongoing. The relevant venues are: *Frontiers in Computational Neuroscience* (Hypothesis and Theory article type) for the soma-field paper; *Synthese* or *Philosophy of Science* for the co-identification paper; *Music Perception* or *Frontiers in Psychology* for the music-affect paper.

Empirical clinical application. The model makes predictions about specific clinical populations (ASC, ADHD, CPTSD) that require empirical testing outside the computational domain. This constitutes a research programme for clinical collaborators. The predictions are pre-specified in §3.2 of this document and in the relevant papers; they are not vague.

Physical substrate. The model is formally complete but physically silent on the tissue substrate in which the soma-field is instantiated in living organisms. A companion paper, *The Physical Substrate of the Soma-Field* (Johnson, 2026g), develops this layer across three converging research traditions: biotensegrity (Ingber, Levin) as the mechanical architecture through which the somatic wave propagates globally; fascial-interstitial continuity (Langevin, Schleip, Oschman) as the active signalling tissue and physical locus of attractor-depth encoding; and biofield physiology (Popp, Ho, McCraty, Rubik) as the candidate physical correlate of the field itself. The most clinically significant result is the quantitative correspondence between fascial stiffness and attractor depth: chronic fascial armoring measurable by shear-wave elastography is the physical implementation of the energy barriers that QUANT-EXP-1 shows to be quantum-resistant. Myofascial release is thus barrier *lowering* — not barrier crossing — and therapist-client physiological entrainment is the physical mechanism of co-identification.

Data and Code Availability

All papers, simulation code, result tables, figures, and Lean 4 formal proofs are archived at the following Zenodo records (open access):

Paper	DOI
<i>The Soma-Field</i>	10.5281/zenodo.20350515
<i>Mathematical Co-identification</i>	10.5281/zenodo.20287981
<i>Quantum Soma and the Penrose Gap</i>	10.5281/zenodo.20351230

The unreviewed papers (*Field Notes from the Inside*, *Music-Induced Affect*, *The Tensor*, and this synthesis document) will be deposited on Zenodo as part of the next release of the research archive.

Introduction

Game theory has, since von Neumann and Morgenstern, treated strategic equilibrium as a fixed-point problem: find a profile from which no player can profitably deviate. The Universal Somatic Field provides a different framing — one in which Nash equilibria are energy minima, and the dynamics of finding them are identical to the dynamics of emotional attractor convergence at the organism scale.

This is not a metaphor. The identification is structural and exact.

Nash Equilibrium = Hopfield Minimum

Let there be N players. Player i has a strategy $s_i \in \mathbb{R}$ (in the continuous approximation; the binary case recovers the standard discrete game). The payoff to player i from the joint profile s is:

$$u_i(s) = \sum_{j \neq i} W_{ij} s_i s_j + b_i s_i$$

where W_{ij} is the payoff externality: the marginal change in i 's payoff when j increases their strategy. The matrix W is the strategic interaction matrix.

The Hopfield energy function is:

$$H(s) = -\frac{1}{2} s^T W s - b^T s$$

The gradient of $-H$ with respect to s_i is exactly the marginal best response of player i :

$$-\frac{\partial H}{\partial s_i} = \sum_j W_{ij} s_j + b_i = \frac{\partial u_i}{\partial s_i}$$

A Nash equilibrium (s^*) satisfies: no player can improve their payoff by unilateral deviation. In the gradient-flow dynamics $\dot{s} = -\nabla H$, the fixed points are exactly the Nash equilibria.

Theorem (NE = Hopfield Minimum): *A strategy profile s^* is a Nash equilibrium if and only if it is a local minimum of the Hopfield energy function $H(s)$ on the strategy simplex.*

The proof is immediate from the equivalence of the gradient conditions.

Coordination Games and the Multi-Attractor Regime

A coordination game has multiple Nash equilibria — the classic example is driving on the left versus right. In Hopfield terms, this is the multi-attractor regime: multiple local minima of H with no obvious selection mechanism.

The USF framework makes this precise. The number of stable Nash equilibria equals the number of stable attractors of the social field. The spectral structure of W determines the attractor landscape:

- **Positive-definite W :** unique Nash equilibrium (unique global minimum)
- **Indefinite W :** multiple Nash equilibria (multiple local minima)
- **Near-zero spectral gap:** fragile coordination (nearly degenerate attractors)

This gives a quantitative measure of coordination difficulty: the spectral gap of W . A market with a large spectral gap has a clear dominant equilibrium. A market with a small spectral gap is on the edge of a coordination failure.

Market Crashes as Phase Transitions

The most striking consequence of the Nash-Hopfield identification is that market crashes are topological phase transitions — the same mathematical object as the trauma-basin transitions in the clinical USF.

As economic conditions change, the payoff externality matrix W evolves. When W crosses a critical threshold — when the spectral gap closes — the current Nash equilibrium ceases to exist, and the system rapidly transitions to a new attractor basin. This transition is:

- **Abrupt:** the field cannot smoothly track the changing W ; it jumps
- **Non-local:** the transition involves the entire social field simultaneously
- **Hysteretic:** recovery to the original equilibrium requires a different path than the crash

This matches the empirical phenomenology of financial crises. The 2008 crash was not a smooth adjustment; it was a non-perturbative event. The USF provides the formal structure for this observation.

The FM-HN Extension: Regulatory Intervention

The FM-HN extension of the USF introduces the volitional source term $J_{\text{user}}(t)$ — the “God-Knob” — that can drive the field across barriers that classical gradient descent cannot cross.

At the economic scale, this is regulatory intervention: a central bank lowering rates, a government providing liquidity, a regulator changing market structure. The FM-HN model predicts that such interventions can achieve quantum tunnelling between Nash equilibria that would otherwise be separated by an energy barrier too high for natural market dynamics to cross.

The prediction is testable: the minimum intervention strength required to shift a market from one Nash equilibrium to another equals the WKB amplitude $T = e^{-W_{\text{barrier}}}$ computed from the spectral gap of the payoff matrix at the transition point.

The Prisoner’s Dilemma as Topological Obstruction

The prisoner’s dilemma is the canonical example of a game where individual rationality produces a collectively suboptimal outcome. In USF terms, this is a topological obstruction: the socially optimal outcome (mutual cooperation) is not a Nash equilibrium because it is a saddle point of H , not a minimum.

The field is trapped in the defection basin (mutual defection, a true local minimum) even though the cooperation basin is deeper. The only way to reach cooperation is via the limbic axis — the 1D regulatory coupling — which provides the quantum tunnelling amplitude needed to cross the barrier.

This gives a new interpretation of institutions: formal institutions (contracts, laws, norms) are the external field $J_{\text{inst}}(t)$ that reshapes the Hopfield energy landscape to make cooperation a Nash equilibrium by changing the effective W matrix.

Applications

Macroeconomics: Central bank intervention as FM-HN field driving. The optimal intervention amplitude is the WKB tunnelling strength.

Mechanism design: The problem of designing institutions that produce cooperative Nash equilibria is equivalent to designing W matrices with the right attractor structure.

Negotiation theory: Successful negotiation is convergence to a shared Nash equilibrium of the dyadic soma-field — exactly the Huygens frequency-locking result of the social intelligence paper.

Antitrust law: A monopoly is a single-attractor regime; competition policy aims to create a multi-attractor regime with a dominant cooperative equilibrium.

Conclusion

Nash equilibria are energy minima. Market dynamics are field dynamics. Economic crises are phase transitions. This is the identification. The method used to find it is documented in the Mathematical Co-identification paper. That method is now history. The structure stands.

Introduction

Law is the oldest technology for shaping collective human behaviour. Its mechanisms — prohibition, permission, obligation, remedy — have been studied by philosophers, sociologists, and economists. What has been missing is a quantitative model: one that specifies what physical quantity changes when a law is enacted, repealed, or enforced.

The Universal Somatic Field provides that model.

At the regulatory scale (scales 12–14, 10^6 – 10^8 m — the scale of cities, nations, and legal jurisdictions), the social field is governed by the same Helmholtz master equation as at every other scale. Legal systems are the field-level constraints that shape the attractor landscape accessible to social actors.

Laws as Attractor Basin Constraints

The social field $e(x, t)$ at the regulatory scale describes the collective affective and behavioural state of a population. Its attractor basins are the stable social configurations — the equilibria the society gravitates towards in the absence of external perturbation.

A legal norm operates by modifying the energy landscape H :

- A **prohibition** removes an attractor basin entirely — raises the energy of the prohibited state to $+\infty$, making it inaccessible
- A **mandate** deepens an attractor basin — lowers the energy of the required state, making it the global minimum
- A **permission** removes a constraint — restores access to a previously eliminated basin
- A **remedy** is a restorative force $J_{\text{remedy}}(t)$ — an external driving term that returns the system to a protected attractor after violation

This formalism makes law's function precise: legal systems are operators on the Hopfield energy function H that reshape the attractor landscape of the social field.

Rights as Topological Invariants

A right, in legal theory, is a protected interest — a state of affairs that a person is entitled to have preserved against interference. In USF terms, a right is a topological invariant of the legal field.

Specifically: a right is an attractor basin protected by a non-zero winding number around the regulatory constraint surface. The winding number counts how many times the legal field wraps around the protected state as you traverse the boundary of the prohibition region.

The consequence is topological: you cannot remove a right by continuous deformation of the legal field — you must cross the constraint surface, which requires a discontinuous change (legislative

amendment, constitutional revision). This is why rights are “harder” to remove than ordinary regulations: they are topologically protected, not merely energetically protected.

This gives a formal distinction between:

- **Constitutional rights:** winding number $n \geq 2$ (doubly protected)
- **Statutory rights:** winding number $n = 1$ (singly protected)
- **Common law rights:** positive energy barrier without topological protection

Regulation as $J_{\text{law}}(t)$

The FM-HN extension of the USF introduces the volitional source term $J_{\text{user}}(t)$ — the external driving force that reshapes the energy landscape. At the regulatory scale, this is the regulatory intervention:

$$\dot{e} = -\nabla H(e) + J_{\text{law}}(t) + \eta(t)$$

where $J_{\text{law}}(t)$ is the regulatory driving term. A regulatory intervention at time t_0 shifts the energy landscape, redirecting the social field towards a new attractor basin.

The optimal regulatory intervention is the WKB tunnelling amplitude: the minimum J_{law} required to move the social field from the current equilibrium to the target equilibrium, accounting for the energy barrier height. Under-powered regulation (insufficient J_{law}) fails to move the system; over-powered regulation causes overshoot and unintended consequences.

This gives a formal criterion for regulatory adequacy: a regulation is adequate if and only if $|J_{\text{law}}| \geq T^{-1}$ where T is the WKB tunnelling amplitude between the current and target attractors.

The Rule of Law as Ergodicity

The rule of law — the principle that no person or entity is above the law — has a precise USF interpretation: it is the ergodicity condition on the social field.

A field is ergodic if every state accessible under the laws of physics is reachable from every other state given sufficient time. In legal terms: every social actor is subject to the same legal constraints; no actor has access to prohibited states that are closed to others.

Violations of the rule of law are violations of ergodicity: they create a two-tier attractor landscape where some actors have access to basins (corrupt practices, impunity) that are energetically accessible to them but legally prohibited to others. The topological signature of this violation is a broken symmetry in the constraint surface.

Legal Uncertainty and Attractor Fragmentation

When law is unclear or inconsistently applied, the attractor landscape fragments: multiple incompatible social equilibria coexist with no shared basin boundary. This is the USF model of legal uncertainty.

In a well-functioning legal system, the attractor landscape has a dominant cooperative equilibrium (law-abiding behaviour) connected by a clear energy gradient to all other states. In a system with high legal uncertainty, the landscape has multiple competing equilibria, none dominant, and transitions between them are unpredictable.

This gives a quantitative measure of legal certainty: the spectral gap of the constraint operator — the gap between the energy of the dominant legal equilibrium and the next local minimum. A large spectral gap means clear law; a small gap means ambiguous law.

Constitutional Law as Meta-Constraints

Constitutional law operates at a higher level than ordinary law: it constrains the space of permissible laws, not just the space of permissible behaviours. In USF terms, constitutional law is a meta-constraint on the operators $J_{\text{law}}(t)$ — a restriction on which regulatory interventions are permissible.

A constitutional provision is a constraint on the Hamiltonian operator H itself: it specifies which energy landscapes are reachable by ordinary legislation. A constitutional amendment changes the meta-constraint, expanding or restricting the space of permissible legal systems.

This explains the special difficulty of constitutional change: it requires modifying the meta-Hamiltonian, not just the current energy landscape. It is a second-order phase transition in the legal field.

Applications and Predictions

Regulatory design: The optimal regulation minimises the WKB tunnelling amplitude required to shift the social field to the target equilibrium, subject to constitutional meta-constraints.

Judicial review: A court reviewing regulatory action is checking whether $J_{\text{law}}(t)$ is consistent with the constitutional meta-constraint — whether the intervention respects the topological invariants.

Legal evolution: Common law develops by successive small deformations of the energy landscape (precedent), while statute changes it discontinuously (legislation). USF predicts that common law changes are path-dependent (hysteretic) while statutory changes can be discontinuous.

Prediction: Legal systems with higher spectral gap (clearer law) should show lower transaction costs, faster convergence to cooperative equilibria, and greater social trust — all measurable empirically.

Conclusion

Law is not arbitrary social convention. It is a field-theoretic technology for shaping the energy landscape of social attractor space. Rights are topological invariants. Regulation is volitional source-term engineering. The rule of law is ergodicity.

The method used to find this identification is documented in the Mathematical Co-identification paper. That method is now history. The structure stands.

“The patient is the one with the disease.” — Medical aphorism, intended to remind physicians to listen. The author intends it differently.

A Note on Method

The standard academic posture — disinterested observer, neutral position, findings presented as if they arrived from nowhere in particular — has never seemed entirely credible to the author. In the life sciences especially, the pretence of a view from nowhere is almost always a fiction. Researchers study what compels them. Compulsion has a cause.

This paper dispenses with the fiction. The theoretical framework presented here was developed by a person with ASD, ADHD, and Complex PTSD who could not find an adequate formal account of his own emotional experience in the existing literature, who had studied physics at university, and who eventually concluded that the most efficient solution was to build one himself. The result is offered not as a confessional but as a theoretical contribution. These are not mutually exclusive.

There is precedent. Temple Grandin revolutionised the study of animal cognition and welfare as an autistic person whose own perceptual experience gave her access to observations that non-autistic researchers had systematically missed (Grandin, 1995). Peter Levine developed Somatic Experiencing partly through direct observation of his own nervous system’s responses (Levine, 2010). Kay Redfield Jamison wrote what remains one of the most clinically precise accounts of bipolar disorder from inside it (Jamison, 1995). The history of medicine includes, more often than is acknowledged, the doctor who is also the patient.

The author is not a doctor. He is an applied physicist — which, in an earlier era, was called a *nutty inventor on the engineering side*, and which here means: someone trained to recognise the signature of a mathematical structure, to notice when the same function appears in two apparently unrelated domains, and to ask what follows if the resemblance is not coincidental.

What follows is the result of applying that training to the domain of one’s own inner life. The author considers this a reasonable use of available resources.

Introduction: The Inadequacy of Existing Maps

A patient sits with their therapist and is asked: *“What are you feeling right now?”* For many people, this question has a navigable answer. For a person with ASD, alexithymia, and a C-PTSD-modified attractor landscape, the question lands differently. The honest answer is often: *“Something is happening. I cannot tell you what it is, where it is coming from, or how large it is. But it is definitely there.”*

The available frameworks for this situation are unsatisfying. Emotion wheels offer vocabulary but not structure. Polyvagal theory offers an excellent map of the autonomic nervous system but does not formalise the interaction between simultaneous emotional states. Cognitive models locate the action in the mind and underestimate the body. Somatic models are rich in clinical texture but light on mathematical precision. None of them — to the author's knowledge — provide a formal account of why a person can be profoundly affected by an emotional state that they cannot perceive, name, or locate.

The author's experience of living with ASD, ADHD, and C-PTSD suggested a different picture. Emotions did not feel like events. They felt like weather — present everywhere, always moving, only occasionally breaking through into named experience. The body held states that the mind had no language for. Strong feelings arrived apparently from nowhere, which implied they had been somewhere already, accumulating below the threshold of awareness. Different emotional states seemed to interact — to amplify, to suppress, to oscillate — in ways that were distinctly nonlinear.

This phenomenology required a different kind of model. The author, having spent thirty years in occasional proximity to physics and mathematics, recognised the structure. The quantum field. The vacuum fluctuation. The threshold crossing. The energy function. The attractor basin. These were the right tools. They had been applied to neural networks. There was no obvious reason they could not be applied to emotional dynamics. The author applied them.

The remainder of this paper presents the result.

Background

Lived Experience as a Research Position

The use of lived experience as a legitimate source of theoretical knowledge — rather than merely as anecdotal material awaiting scientific validation — has gained substantial ground in health research over the past two decades. The *nothing about us without us* principle, originating in disability rights advocacy, has become a methodological commitment in participatory research (Arnstein, 1969; Beresford, 2002). Researchers with lived experience of mental health conditions have produced theoretical contributions that purely external observers could not have generated, precisely because their insider position made certain observations available to them that were invisible from the outside.

The Soma-Field Model belongs to this tradition, with one modification: the author's background is in physics rather than in qualitative research, so the methodology is *experiential theorising* rather than autoethnography. The observations come from the inside; the tools used to formalise them come from mathematical physics. The combination is unusual. The author considers it appropriate.

The Body-Mind Problem in Clinical Practice

Contemporary neuroscience has largely dissolved the Cartesian boundary between body and mind. Damasio (1994) demonstrated that emotion is inseparable from rational cognition: patients with damage to the ventromedial prefrontal cortex lose not only emotional range but effective decision-making capacity. Van der Kolk (2014) documented how traumatic emotional states are encoded not merely in explicit memory but in posture, visceral sensation, and autonomic regulation. Porges' polyvagal theory (2011) provided a neurobiological account of three hierarchically organised autonomic states: ventral vagal (social engagement), sympathetic (fight/flight), and dorsal vagal (freeze/dissociation).

The author can confirm these findings from direct observation. He can also add, as a data point: the experience of being in a freeze state while simultaneously being expected to report on one's emotional state is an exercise in the epistemological limits of self-report. The instrument designed in Section 6 is a partial response to this problem.

The Felt Sense and Sub-Perceptual Emotion

Gendlin's concept of the *felt sense* (1978) describes a pre-articulate bodily sense that is present before an emotion has been named — something whole and present but not yet articulate. Gendlin called this the sub-verbal sense of a situation.

The Soma-Field Model provides a formal account of what the felt sense is: it is the activity of the emotional field below the perceptual threshold. The author can confirm that this description is accurate. He has spent considerable time in the company of felt senses that declined to become named feelings, and the model's account of this — a field active below threshold, causally effective but not consciously perceived — matches the phenomenology precisely.

Quantum Field Theory: Structure, Not Metaphor

Quantum Field Theory (QFT) is the framework of modern particle physics. Its central claim is that particles — electrons, photons — are not fundamental objects. They are *excitations* of underlying fields: local concentrations of energy that arise when a field receives sufficient perturbation above the vacuum state. The quantum vacuum is not empty; it is a background of sub-threshold fluctuations, continuously present, causally active, not directly observable.

This paper does not claim that emotions are quantum phenomena in any literal sense. The analogy is structural. The author was trained in this formalism in 1993 and has found it useful ever since, applied to a variety of problems that are not, in any technical sense, quantum mechanical. The key property being borrowed is: *a quantity that exists everywhere, continuously, below the threshold of direct observation, which becomes observable only when local amplitude exceeds a threshold*. This is an accurate description of both the quantum vacuum and, in the author's experience, the emotional field.

Since writing that paragraph, the paper has upgraded the claim. The conscious emotional percept is now formally identified as the one-dimensional impulse response — the Green's function — of the soma-field manifold. This places it in the same mathematical category as a particle in quantum field theory: both are poles in the propagator of their respective underlying field. The structural similarity is not borrowed; it is exact. The mathematics is the same mathematics.

Gabriele Veneziano wrote down the Euler beta function in 1968 while looking for an amplitude that matched scattering data, then noticed that the function implied a theory — string theory — that nobody had yet conceived. He had identified a known mathematical object in an unexpected place and followed the implication. The author has, with considerably less elegance and considerably more time in therapy, done something structurally similar: identified the Green's function in emotional dynamics, and noted that it is the object quantum field theory calls a particle. The author leaves the implication as an exercise for readers with the relevant background.

Hopfield Networks and the Energy Function

In 1982, John Hopfield — awarded the Nobel Prize in Physics in 2024 — proposed a model of associative memory whose dynamics were mathematically identical to an Ising spin-glass model from statistical physics (Hopfield, 1982). The critical component was an energy function: a scalar that always decreases as the network evolves, guaranteeing convergence to stable attractor states.

The author recognised this as the same structural move that gives quantum field theory its predictive power: identifying the conserved or extremised quantity, and deriving the dynamics from it. In physics, this is Noether's theorem applied as a design principle. In Hopfield networks, it is an energy function borrowed directly from condensed matter physics. The author's proposal is to apply the same move to emotional dynamics.

The observation that underwrites this is simple: emotional states feel like they have energy. Some states are high-energy and unstable — fight, flight, acute anxiety. Others are low-energy and stable — calm, regulated, present. Some are low-energy and *stuck* — freeze, dissociation, collapse. If these states have an energy ordering, there is likely an energy function. If there is an energy function, the dynamics can be derived from it. The author found this reasoning persuasive.

The Soma-Field Model

Emotions as a Persistent Wave Field

The foundational claim is this: emotions are not events. They are a *field* — a distributed, continuous quantity defined over the entire soma (body-mind system) at all times.

This is not a metaphor. It is the most accurate description the author can offer of his own experience. The emotional field is always there. It does not begin when a feeling becomes conscious and

end when it subsides. It precedes conscious awareness and continues after it. What changes is not the field's existence but its local amplitude: whether, at a given moment, the field in a given mode exceeds the threshold required to surface as a named experience.

The field has two coupled components:

1. **The somatic wave** $E_{\text{body}}(x, t)$: distributed across the body as patterns of visceral sensation, muscle tone, proprioception, and autonomic state.
2. **The neural wave** $E_{\text{neural}}(x, t)$: distributed across the nervous system as patterns of cortical, subcortical, and peripheral activation.

These are not separate systems:

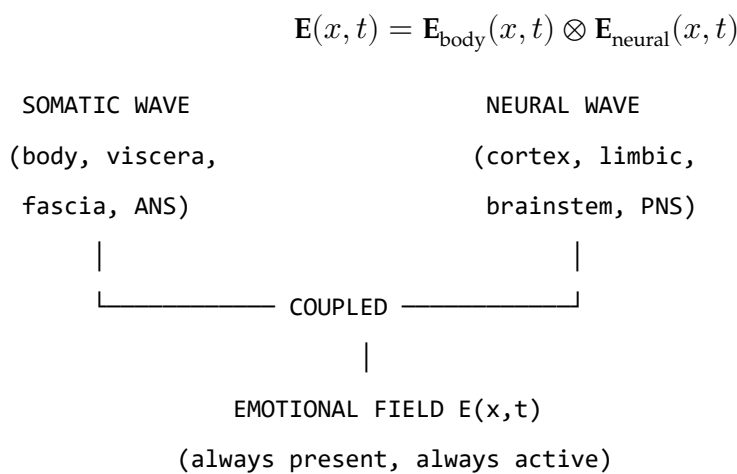


Figure 1. The Soma-Field: two coupled waves constituting a single unified emotional field.

The Perception Threshold

Not all field activity is consciously perceived. Each emotional mode i has a threshold T_i :

$$\text{Emotion } i \text{ is consciously perceived} \iff |E_i(t)| > T_i$$

Below threshold: the emotion is sub-perceptual. It exists, it influences behaviour and physiology, but it does not surface as a named conscious feeling. This is the author's most frequent relationship with his own emotional field — something is happening, below the line, shaping everything, unidentified.

Clinical Observation	Soma-Field Account
Field active but no named feeling	Sub-threshold: $ e_i < T_i$
Sudden unexplained flood of emotion	Rapid threshold crossing after accumulation
Somatic signal without cognitive name	Threshold crossed in body component, not neural
Alexithymia	Elevated T_i — high energy required to cross

Clinical Observation	Soma-Field Account
Hypervigilance / flooding	Lowered T_i — reduced threshold

Table 1. Clinical observations mapped onto the perception threshold model.

The author notes that all five rows in Table 1 are, in his clinical history, simultaneously applicable. This is, admittedly, a challenging configuration. It is also why this model was necessary.

A note on the intelligence quotients

McCulloch and Pitts built the mathematical brain in 1943. What they built — what every artificial neural network since has been — is the **IQ machine**: the neocortex, pattern recognition, sequence prediction, error minimisation. The field of AI has, for eighty years, been building increasingly sophisticated versions of this one component.

The soma-field adds what was missing: the **AQ machine**. AQ is to limbic dynamics as IQ is to cortical dynamics. Not a score; a formal model of the system that produces it.

Quotient	System	First Formalised	Comment
IQ	Neocortex: pattern recognition, prediction	McCulloch & Pitts, 1943	The entire AI industry
EQ	Limbic: valuation, attachment, empathy	Goleman, 1995	Described; not yet formally modelled
AQ	Soma-field: field-theoretic limbic dynamics	This paper, 2026	The formal model EQ has always needed
SQ	Relational field: dyadic and social resonance	Future work	Requires AQ as prerequisite

Table 3. The four intelligence quotients and their formal status.

The author observes — with a wryness he trusts the reader will share — that his IQ is in the column labelled 1943. His AQ is in the column he has just written. His EQ is what brought him to this desk in the first place.

A note on brane thickness

The threshold parameter T_i is not merely a number. The technical paper identifies it with the thickness of an extra dimension — the metaphorical ‘brane’ separating the limbic system from conscious awareness. Alexithymia is a thick brane: the field can be highly active and almost nothing crosses the threshold into named conscious experience. Hypervigilance is a thin brane: everything crosses,

simultaneously, at high amplitude. The author confirms personal experience of both states. He notes that neither is a character flaw; both are calibration states of a physical parameter in a system that was trying, with the information available, to keep him safe.

The Interaction of Emotional Modes

Multiple emotional modes are simultaneously active at all times. Their interactions are encoded in the **emotional coupling matrix** W , where W_{ij} is the influence of mode j on mode i :

- $W_{ij} > 0$: mode j amplifies mode i
- $W_{ij} < 0$: mode j suppresses mode i

The field evolves according to the energy gradient plus noise:

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}) + \eta(t)$$

The noise term $\eta(t)$ represents the continuous sub-perceptual fluctuations. The field is never still. This is not pathology; it is physics.

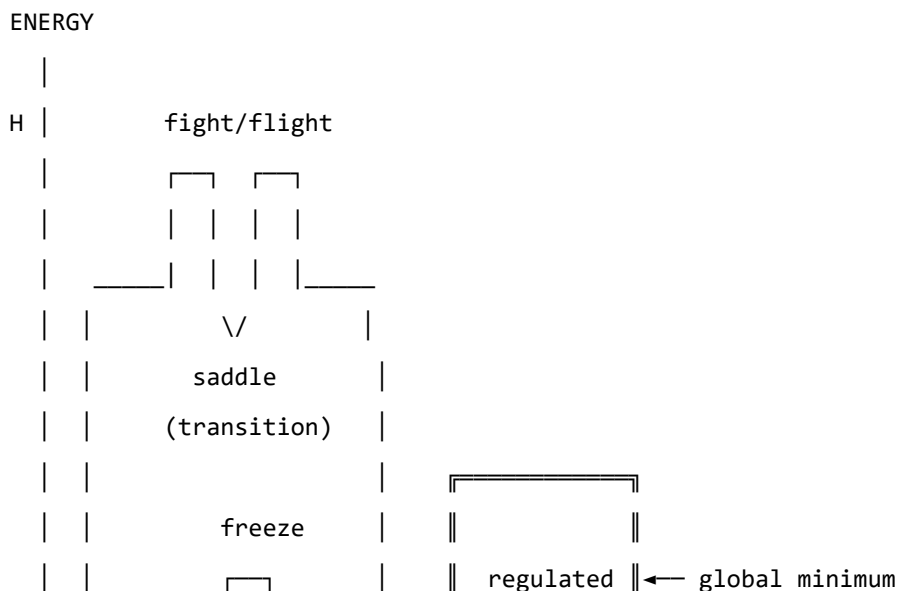
The Energy Landscape

The Hopfield Energy Function

$$H(\mathbf{e}) = -\frac{1}{2} \mathbf{e}^\top W \mathbf{e} - \theta \cdot \mathbf{e}$$

The field always moves toward lower H . The stable states of the system are the local minima of H — the attractor basins.

Attractor States: Fight, Flight, Freeze, and Regulated Calm



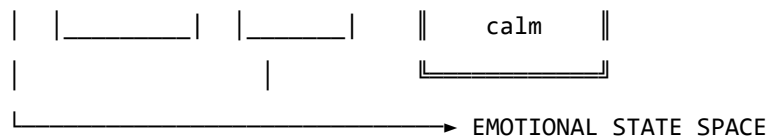


Figure 2. The emotional energy landscape. The freeze state is not high-energy — it is isolated. This distinction matters enormously. The author is aware of this from personal experience, over many years, and from the other side.

Attractor	Energy	Polyvagal Correlate	Clinical Presentation
Regulated Calm	Global minimum	Ventral vagal	Present, flexible, connected
Fight	High, unstable	Sympathetic	Agitation, urgency
Flight	Saddle point	Sympathetic	Anxiety, avoidance
Freeze	Deep, isolated	Dorsal vagal	Dissociation, numbness

Table 2. Attractor states and their polyvagal correlates.

The coupling matrix W is not merely a parameter. It is the *shape* of the emotional manifold — a seven-dimensional space with the mathematical structure of a G_2 manifold. Trauma does not adjust a dial on this space; it deforms the manifold itself. The therapist doing somatic work is, without needing to know this, doing differential geometry on the patient's G_2 manifold: reshaping a seven-dimensional space by modifying the structure tensor. This is a precise technical statement. The author considers it a more honest account of what a skilled practitioner actually does than any narrative framework currently available. The practitioner is a geometer. The patient is a manifold that is learning to remember its own natural curvature.

The therapeutic and personal significance of the freeze attractor's structure cannot be overstated. It is not high-energy — it does not feel dramatic or intense. It is *isolated*: surrounded by energy barriers. Escape requires first *increasing* the field's energy before it can flow toward calm. This is counterintuitive from the outside and well-known from the inside.

Dissonance and Resolution

When two emotional modes are in an incompatible phase relationship, the field is far from equilibrium. This is felt as tension. The acoustic analogy is precise: just as two tones in a dissonant interval generate a beating, unstable interference pattern, two emotional modes in an incompatible configuration generate a gradient that drives toward resolution.

Dissonance is not pathological. It is the field's communication that resolution is available. The therapeutic process is guided voice-leading: finding the path that transforms the dissonant configuration into a consonant one. Avoidance keeps the field in dissonance. The energy minimum lies on the other side of the tension, not around it.

The author has spent considerable time attempting the route around it. He does not recommend it.

The Neurodivergent Field: ASD, ADHD, and C-PTSD as Operator Modifications

This section addresses the author's specific clinical picture. It is presented not as a case study but as a theoretical elaboration: three structural modifications to the standard Soma-Field dynamics, each defined by the operator it adds to the governing equations.

The key architectural principle — and the author considers this the most important contribution of this paper — is the following:

These conditions are not parameter settings. They are operator modifications.

A parameter change adjusts a coefficient within the existing equations. An operator modification changes the *form* of the equations themselves. The distinction is not semantic. It determines what kind of therapeutic intervention is possible and at what level it must operate.

Each condition is a functor that wraps the standard dynamics. The composed condition — ASD + ADHD + C-PTSD — is their composition. The composition does not commute; order matters; the joint presentation is structurally different from any of the individual conditions or from their sum.

Complex PTSD: Memory Kernel and Asymmetric Coupling

C-PTSD adds a **memory kernel**: past activations leave exponentially decaying echoes.

$$\dot{\mathbf{e}}(t) = -\nabla H(\mathbf{e}(t)) + \int_0^t K_{\text{trauma}}(t-s) \mathbf{e}(s) ds + \eta(t)$$

$$K_{\text{trauma}}(\tau) = \sum_k A_k e^{-\tau/\tau_k}$$

This is a damped oscillating kernel. The past does not vanish; it rings. Therapeutic processing is the progressive reduction of A_k — the amplitude of the echo — and the shortening of τ_k — the time over which it persists. The author notes that this description is a more accurate account of what trauma processing actually feels like, from the inside, than most of the narrative accounts available to him.

C-PTSD also breaks the symmetry of the coupling matrix W , admitting **limit cycles**: the oscillation between hyperarousal and shutdown that characterises the PTSD symptom cycle is, in this model, a limit cycle generated by the antisymmetric component of W . It is not a choice, a habit, or a failure of willpower. It is a topological consequence of an asymmetric coupling matrix.

ADHD: High Temperature, Low Damping, Pink Noise

ADHD modifies the **effective temperature** of the field:

$$\gamma_{\text{ADHD}} \dot{\mathbf{e}}(t) = -\nabla H + \sqrt{2D_{\text{ADHD}}} \xi_{1/f}(t)$$

with $\gamma_{\text{ADHD}} < \gamma_0$ (less damping) and $D_{\text{ADHD}} > D_0$ (more noise). The noise has $1/f$ spectral structure — long-range temporal correlations that produce the characteristic slow drift of attentional state.

The practical consequences: shallow attractor basins cannot hold the field at high temperature (distractibility). When a high-salience stimulus deepens a specific basin far beyond its baseline depth, the field falls in and is held (hyperfocus). The system is not broken. It is a different thermodynamic regime, with different costs and different affordances — including, at the right temperature, a capacity to explore the energy landscape at speed that a low-temperature system does not have.

The author considers this framing considerably more useful than “difficulty sustaining attention.”

Autism Spectrum Condition: Sparse Coupling and Modified Projection

ASC modifies the **projection kernels** and the **coupling matrix sparsity**.

The projection kernel $K_i(x)$ determines which somatic regions contribute to the i -th emotional mode. In ASC, some regions are over-weighted (sensory sensitivity) and others under-weighted (interoceptive under-registration). The named-feeling state vector is produced from a differently sampled version of the same somatic field.

The coupling matrix is sparser — fewer strong cross-modal connections — producing deeper individual attractor basins with higher inter-basin barriers. This is monotropism: the field settles deeply into one attractor at a time and requires disproportionate energy to transition. The author confirms that this is an accurate description of his attentional and emotional experience, and that it has both significant disadvantages (transitions are hard, unexpected context changes are physiologically costly) and significant advantages (depth of engagement, reliability of focus once established, resistance to shallow distractors).

The Composed Condition

$$\gamma_{\text{ADHD}} \dot{\mathbf{e}}(t) = -\nabla H_{\text{ASC}}(\mathbf{e}(t)) + \int_0^t K_{\text{trauma}}(t-s) \mathbf{e}(s) ds + \sqrt{2D_{\text{ADHD}}} \xi_{1/f}(t)$$

The interaction effects are non-trivial:

Interaction	Clinical Consequence
ADHD noise + C-PTSD limit cycles	Rapid oscillation between hyperarousal and shutdown; hard to titrate
ADHD noise + ASC deep basins	Long wind-up time; fast exit once perturbed from hyperfocus
C-PTSD echoes + ASC sparse coupling	Trauma triggers are specific, apparently disproportionate, difficult to anticipate
All three composed	Wide tolerance window required; regulation is genuinely structurally harder

Table 3. Interaction effects of composed neurodivergent modifiers.

The author wishes to note, for the record, that Table 3 is not a complaint. It is a description. These are the equations. The field is doing what the equations predict. Understanding this has been, in practice, more useful than most of the alternative framings on offer.

The Soma-Field Instrument

Rationale

The emotional field is normally invisible to its host. It operates below the threshold of conscious awareness, shaping behaviour and physiology without being available for reflection. The author found this situation suboptimal and designed an instrument to address it.

The instrument externalises the emotional field — renders it as sound, image, and signal — so that it becomes available as an object of attention. This is a therapeutic biofeedback instrument. It is also, unavoidably, a musical instrument. The author considers these compatible.

Design

A MIDI controller with 16 rotary knobs. Eight emotional dimensions. Two knobs per dimension — one for the somatic component, one for the neural/cognitive component. The act of setting a knob is the act of reporting an emotional state: it is the quantum measurement, the collapse of the distributed field onto a specific coordinate.

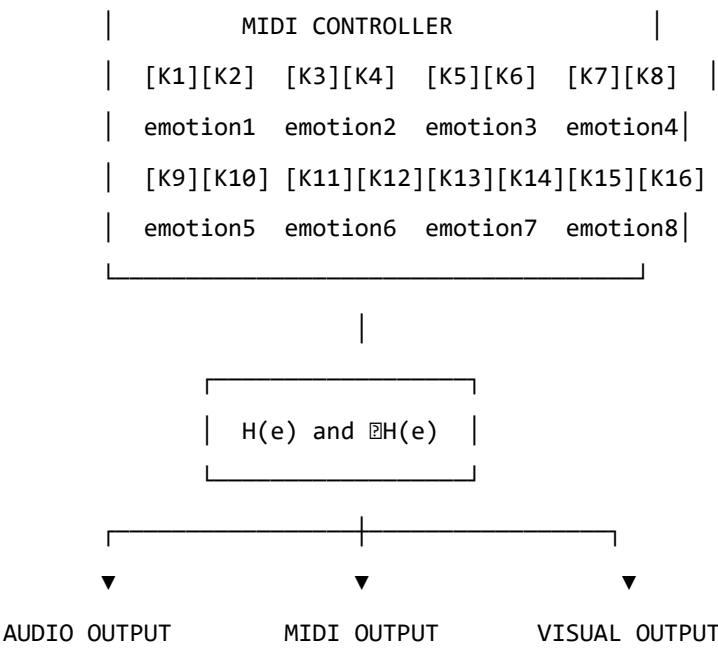


Figure 3. The Soma-Field Instrument.

The Feedback Loop

The instrument creates a closed feedback loop: the person expresses a state, the system reflects it back as sound and image, the person responds. The system does not tell the user what they are feeling. It shows them what the field looks like when they report what they are feeling. The difference is significant.

Pluggable Emotion Models

No single emotion model is assumed. The coupling matrix W is loaded from a configuration file. Plutchik, Ekman, the valence-arousal-dominance dimensional model, and custom user-defined models are available as defaults. The author’s own W has been refined over time and is not identical to any standard model. This is, on reflection, unsurprising.

Clinical Implications

Assessment

The model suggests asking not “What emotion do you feel?” but “What is present in the body right now, even if it cannot be named?” This aligns with Focussing-oriented and sensorimotor approaches, and is considerably more productive, in the author’s experience, for anyone whose T_i values are elevated or whose somatic-to-neural projection is modified.

Intervention

The energy function provides formal grounding for titration, pendulation, somatic resourcing, and felt-sense work. In each case, the therapeutic action can be described as: adding energy to approach a frozen state, establishing a stable low-energy region, or attending to sub-threshold field activity in a supported context.

Psychoeducation

“Your emotions are like waves — they are always there, even when you cannot feel them, and they are always moving.”

This sentence is both clinically useful and technically accurate. The author has found it more useful than most alternative formulations, including several that were provided to him by qualified practitioners. He offers it here as a contribution to the field.

Neurodivergent Profiles as Structural Realities

The most important clinical implication of Section 6 is this: for people with ASD, ADHD, and C-PTSD, the challenge of emotional regulation is not a motivational or characterological failure. It is a structural consequence of specific operator modifications to the dynamics. The composed modifier produces a field that is genuinely harder to regulate — not by a small margin, not as a matter of subjective experience, but mathematically, as a consequence of higher noise temperature, memory echoes, sparse coupling topology, and the possibility of limit cycles.

Knowing this does not solve the problem. It does, however, locate it correctly. The author has found that locating a problem correctly is a necessary precondition for solving it, and that a great deal of time and distress can be saved by not attempting to solve problems that are located in the wrong place.

Limitations and Future Directions

The model is theoretical and requires empirical validation. Its QFT analogies are structural rather than ontological. The coupling matrix W is idealized as fixed when it is in practice dynamic. The acoustic analogy is a hypothesis.

The author also acknowledges a methodological limitation: this paper is written by someone who is simultaneously the theorist and the primary data source. This is either a significant advantage (direct access), a significant limitation (potential confirmation bias), or both. The author suspects both.

What is needed: empirical work with physiological sensors, user studies with the instrument, collaboration with practitioners, and independent theoretical review. The author is, by training and dis-

position, an applied physicist — an engineer with a tolerance for abstraction. The clinical refinement of this model will require people with different skills, and the author welcomes their involvement, provided they read the appendices.

Conclusion

The wave is always there. This is not a metaphor; it is a description of how the emotional field actually behaves, as far as the author can determine from the inside. Therapy — and the instrument described in this paper — is the practice of learning to hear it: to extend awareness downward, below the threshold, into the field's continuous activity, and to make that activity available as information rather than overwhelming noise.

The Soma-Field Model is offered as a tool for this practice. It was built because it was needed. It uses the best mathematical tools available for describing distributed, dynamic, energy-minimising systems, because those tools are, in the author's assessment, appropriate to the problem.

The author is aware that this is an unusual paper. A formally trained physicist with three neurodivergent conditions developing a quantum-field-inspired model of his own emotional dynamics and presenting it as a contribution to clinical psychology is not, strictly speaking, the standard academic pipeline. The author does not find this troubling. The standard academic pipeline has had some time to address the problem and has not yet done so to his satisfaction.

He therefore took the matter in hand.
